



# SIKORSKY UH-60M BLACKHAWK FOR X-PLANE



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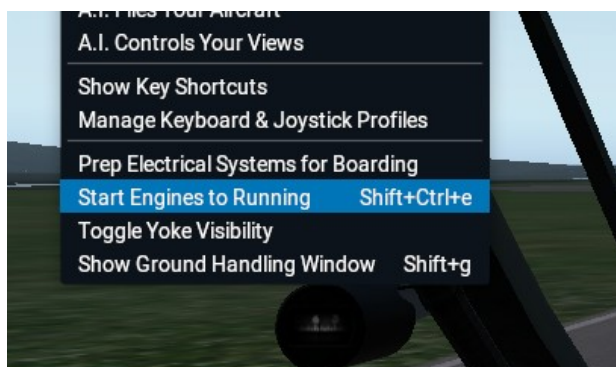
**Missions**

**Datarefs and Commands**



## Quick Start

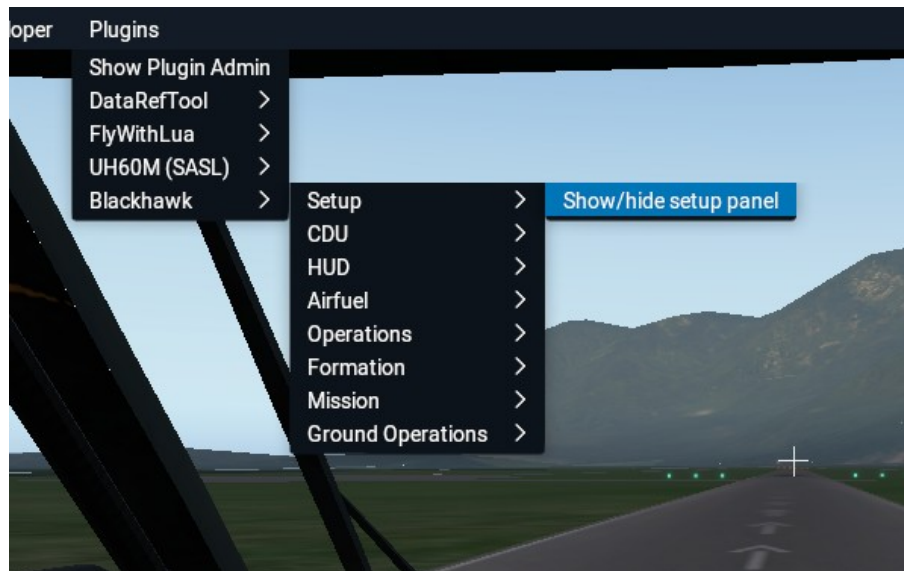
- Make sure you do not have your joystick mapped to a „throttle“ axis. If you have a hardware throttle, use the „wing sweep“ axis instead !
- Start the helicopter either with „running engines“ or use the „autostart“ option from the „Flight“ pull-down menu.





## Configuration

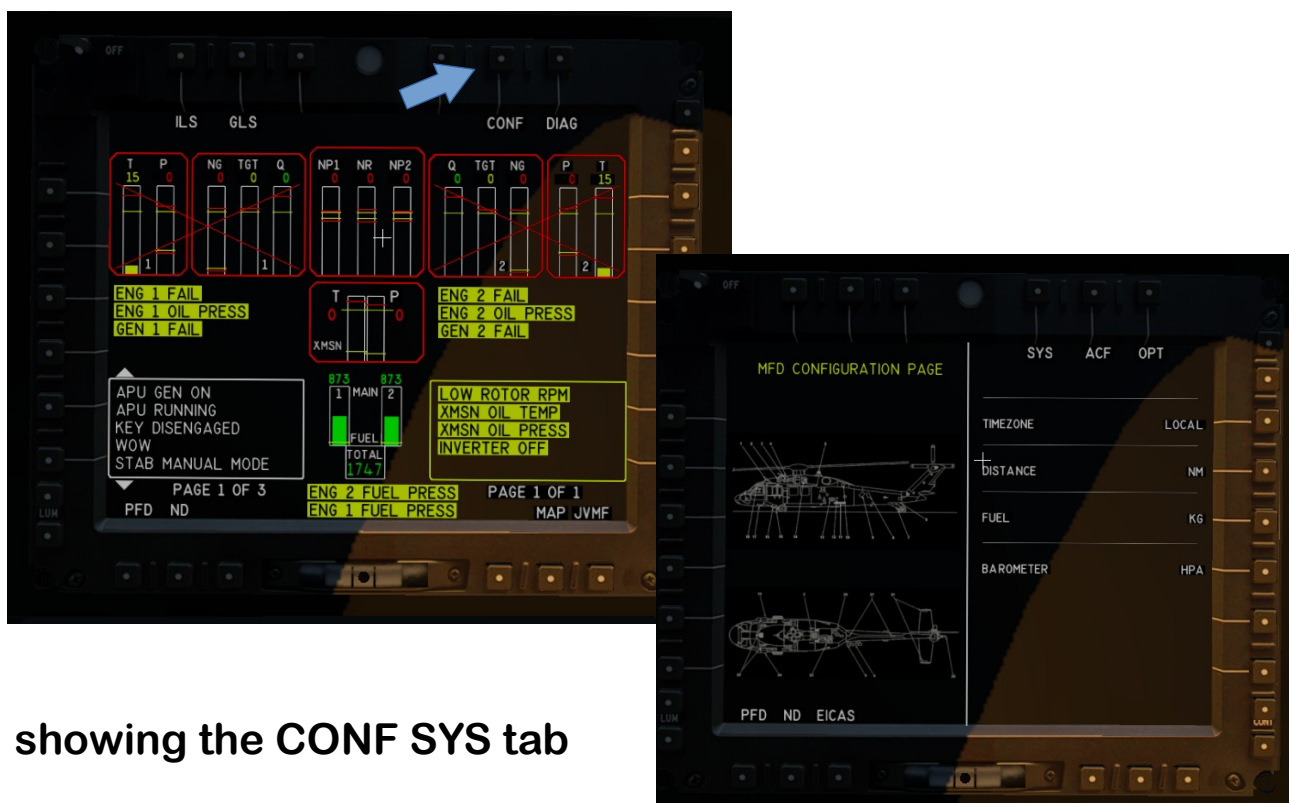
To configure the Blackhawk you can open the „setup panel“ from the „Blackhawk“ setup sub-menu inside the „Plugins“ pull-down menu.



Alternatively you can click on the upper-right-corner of the cockpit dash. If uncertain use the „Show instrument click regions“ under the „View“ menu.

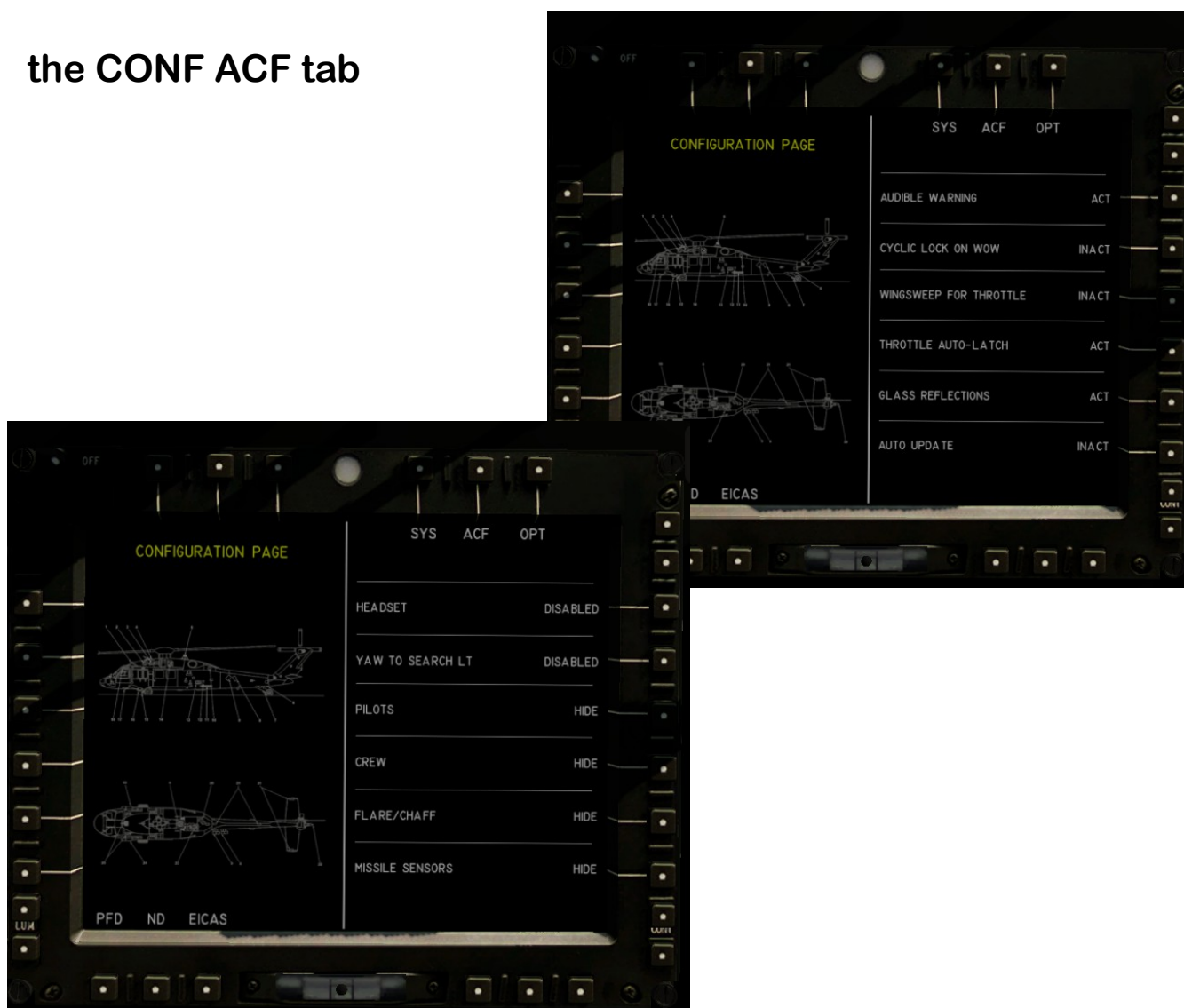


Other configurations can be done from the Multi-Function-Displays (MFD) under the „CONF“ button on the „EICAS“ screen (of course batteries have to be switched ON first).



showing the CONF SYS tab

the CONF ACF tab



and  
the CONF OPT tab

settings and configurations are saved in either the `acf_config.ini` located in the `acf` folder or in the `opt_config.ini` located in the selected livery folder, depending whether they are „global“ or „livery“ related.



## Cold & Dark

On the SETUP panel you can select the CHECKLIST option. A popup will guide you interactively through the steps required to start the helicopter manually from cold&dark.



At the end of the list the engine-start can either be performed by pressing the white buttons on the throttle levers located on the over-head panel, or by assigning the appropriate command to a joystick button ( see “Datarefs and Commands” section ).



Please note that the “Ignition Key” needs to be set to the “ON” position !



Check messages on the EICAS screen and confirm them by pressing the CAUTION button or via the mapped command. Audible alarm can be configured in the CONF menu on the MFD.



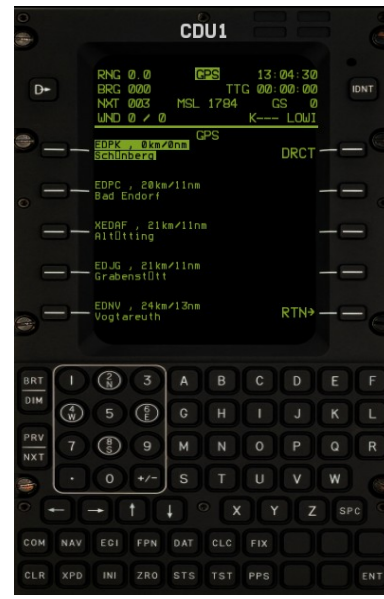


# CDU – Control and Display Unit

## FMS – Flight Management System

The CDU or FMS is being used to control communication and navigation systems, flight plans and missions. A mission is similar to a flight plan but has additional functionality such as running X-Plane commands depending on programmable conditions. More about it in the “Missions” section.





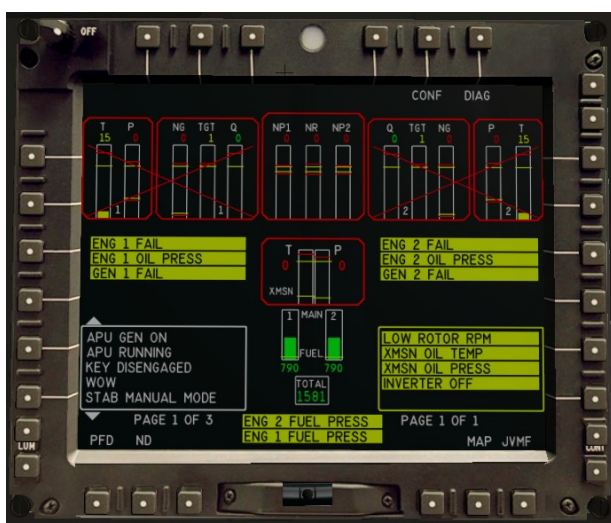
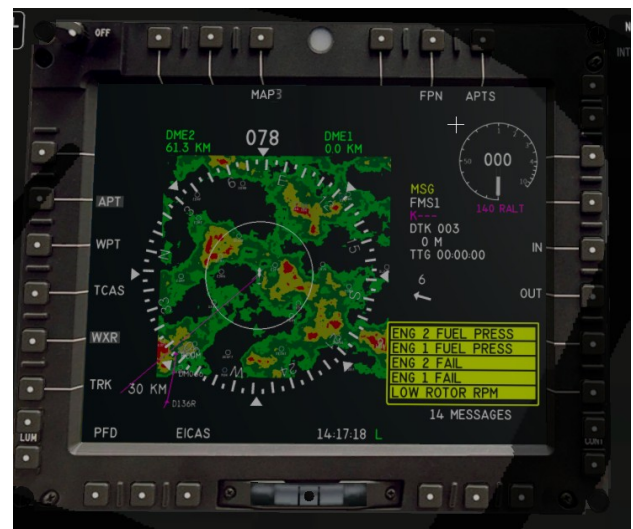
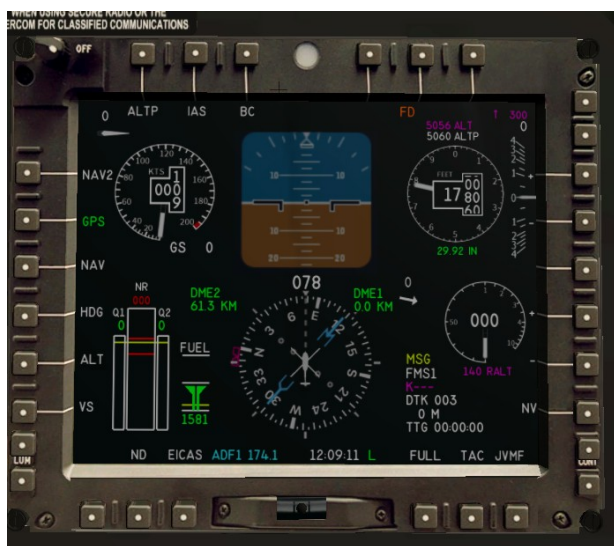


Clicking on the CDU displays in the pedestal will open a CDU popup. This can be helpful in VR. The popups are movable and resizable.

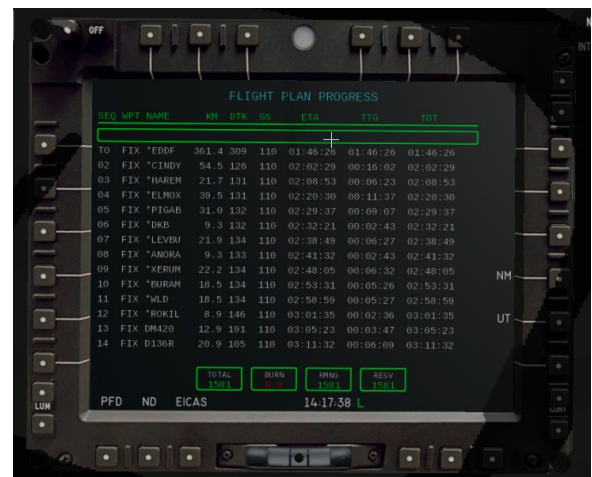
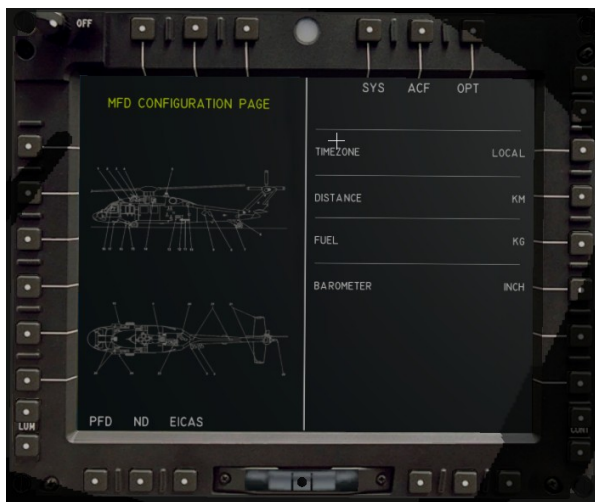




# MFD – Multi Function Displays



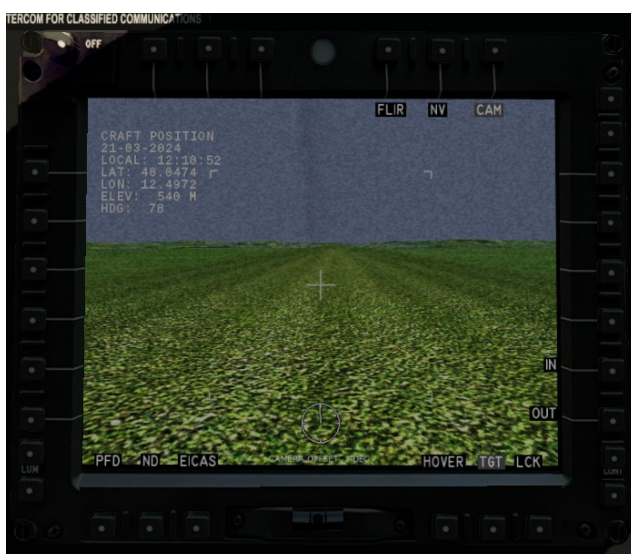




Avitab support (also on iPad)



Camera guided target locking



## JVMF MESSAGES

JVMF stands for Joined Variable Message Format.

It is kind of a military email system. In the simulation it is used to create "tasks" for the crew. Via the "Operations" pull down menu a task to extract a Seal from a random location can be created.

An OSM query in the background tries to find an appropriate spot in the area 20km around the current position. Once the message is being accepted ( ACK = acknowledged ), the FMS is being programmed with the coordinates and so can lead the pilot to the Seal.

The JVMF-softkey switches to **YELLOW** as soon as a new "unread" message shows up in the Inbox. The symbols ">" greater-than" and "?" indicate "un-read" and "un-acknowledged" messages.



See: [https://www.youtube.com/watch?v=2KXv8P6lc\\_w](https://www.youtube.com/watch?v=2KXv8P6lc_w)



## FDCP – Flight Director Control Panel



The FDCP is the control panel for the autopilot. The 5 rotaries for RALT (radio altimeter) , ALTP (programmed altitude). ALT (altitude), IAS (indicated airspeed) and HDG (heading) can be used to change the values in the displays. Pushing the rotaries syncs the display with the actual values (i.e. current altitude).

The black buttons under the displays allow to activate/deactivate the related “upper” AP (autopilot) function. If a function is activated it will be indicated by a green LED inside the button. Some function also have an “armed” state, which is indicated by an amber LED.

The very right button in the upper row controls and show whether the FD (flight director) is engaged/coupled. If it is coupled (CPLD) then the FD controls the aircraft, otherwise the display shows DCPL which stands for de-coupled.

The very left button in the upper row engages RNAV (radio navigation) which means the selected navigation source is providing input to the AP. Valid sources are NAV1, NAV2, LOC1, LOC2 , FMS. They can be selected with the outer ring of the very left rotary in the upper row.

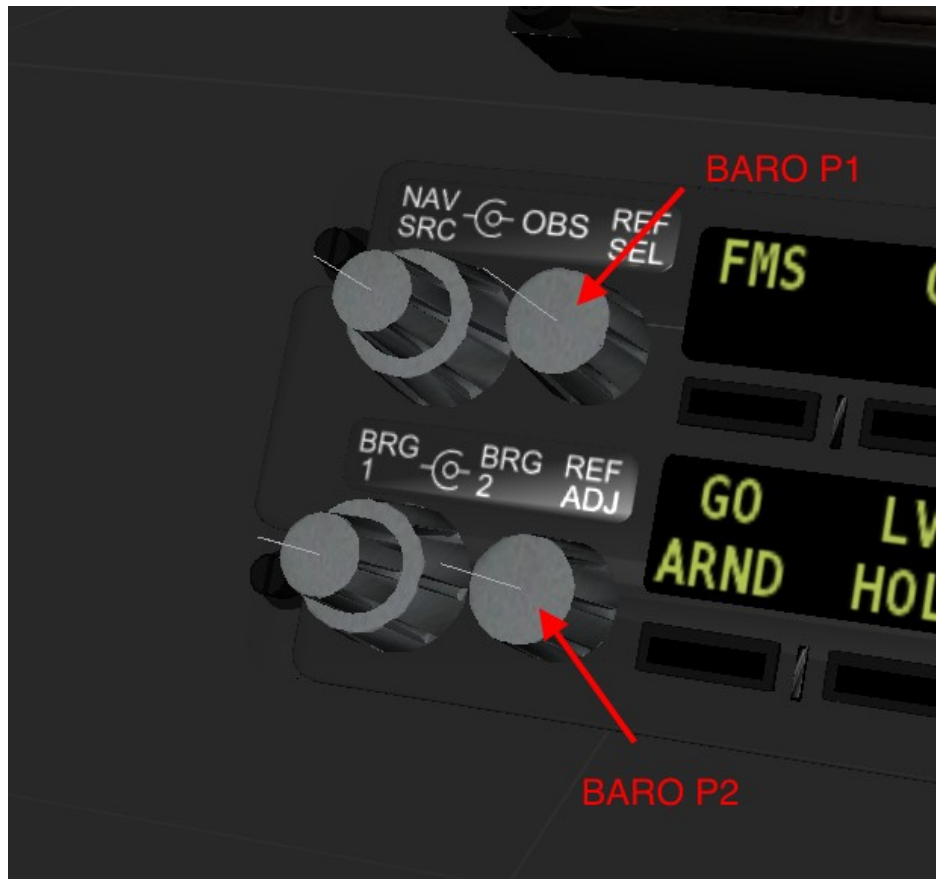
“GO ARND” = GoAround will set the AP to climb with 700 ft/sec and 70 kts.

“LVL HOLD” = LevelHold enables the PID controller which keeps the current altitude-above-ground. It is armed (amber LED) unless vertical speed is less 0.1 m/sec and kias  $\leq$  50 kts. Then it will change to green.

“HVR POS” = HoverHold enables the PID pid controller which keeps the current position if the helicopter hovers with less the 1.5 m/sec horizontal speed.

**NOTE: All AP functions are DISENGAGED if we have WOW (weight-on-wheels). This also applies to all weapons !!**

The pilots and co-pilots barometric pressure can be set using the 2 knobs on the left side of the FDCP



Changing between Inch and hPa can be made in the MFD config page under the SYS tab.

## Cameras

The Blackhawk comes with a built-in multi-camera system. It allows you to do role-playing games.



The system tracks each individual cam position and also allows to save, load and recall position sequences ( i.e. preflight check )

The cameras can be selected through the setup-menu or by using one of the available commands (see DataRefs and Commands).

For example in combination with LVL-HOLD and HVR-HOLD you can observe HRST operations as a Crew Chief while sitting in the Cargo Area.

Hint: make sure you engage your AP before you leave your pilots seat ;)



# Operations

## HRST Fast-Rope-Extraction

Requirement: "HRST" option enabled in setup menu.

Command: uh60m/ops/drop\_dude

Every time the command gets called, a Seal will slide down on one of the ropes and if on the ground, moves to a random "defense position". This also works on moving objects like the aircraft carrier.

## Air-Refuel

Requirement: "REFUEL PROBE" enabled in setup menu.

Command: uh60m/ops/c130\_show.

First command will call the C130. It will adopt to your heading and speed.

Second command will end operation. C130 will climb and disappear.

Third command will delete the C130 immediately.

## Target tracking

Requirement: "FLIR CAM" enabled in setup menu.

On MFD1 select TAC and then TGT. Clicking the mouse into MFD display will show a red aiming mark which allows to control the cam with the mouse. Pressing the LCK button (or use the equiv. command) will lock the camera to the aimed position. A yellow pole will show up at the locked location. A 2<sup>nd</sup> click will release the target and the camera moves back to the default position.

## Extract Seal

An OSM query is being used to select a random location in the range of 20km to place a Seal at. A JVMF order is being created to get him home. If the message is ACKnowledged, the FMS is programmed with the location. Fly to the Seal and land next to him to take him on board.

Note: might not work in areas where no OSM data is available.

# Lights

## Search light

Use command “uh60m/lights/srch\_Lt\_on” to switch search light on and off. Other commands to steer the search light: ‘srch\_Lt\_up’, ‘srch\_Lt\_dn’, ‘srch\_Lt\_lft’ and ‘srch\_Lt\_rgt’. On ground (wow) the search light can be coupled to the yaw axis (pedals).

## Landing light

Use command “uh60m/lights/ldg\_Lt\_on” to switch landing light on and off.



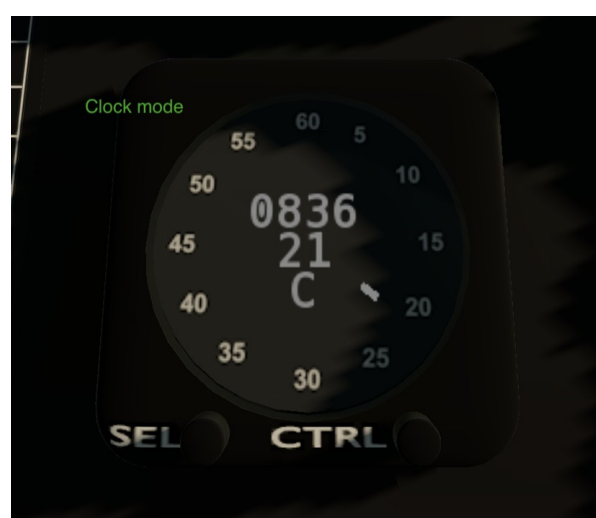
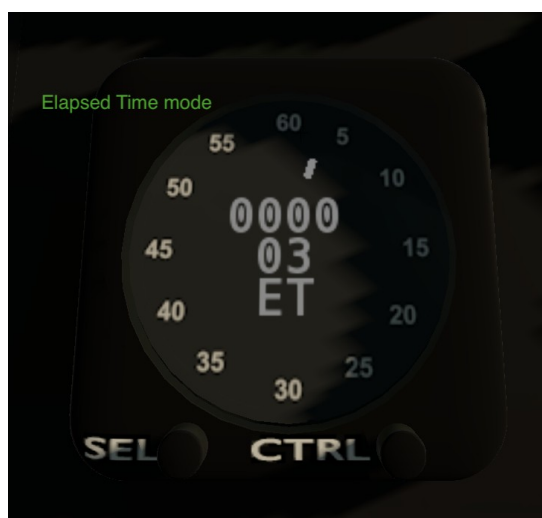
## Instrument lights



## Cockpit, Cabin dome and external lights



## Clock

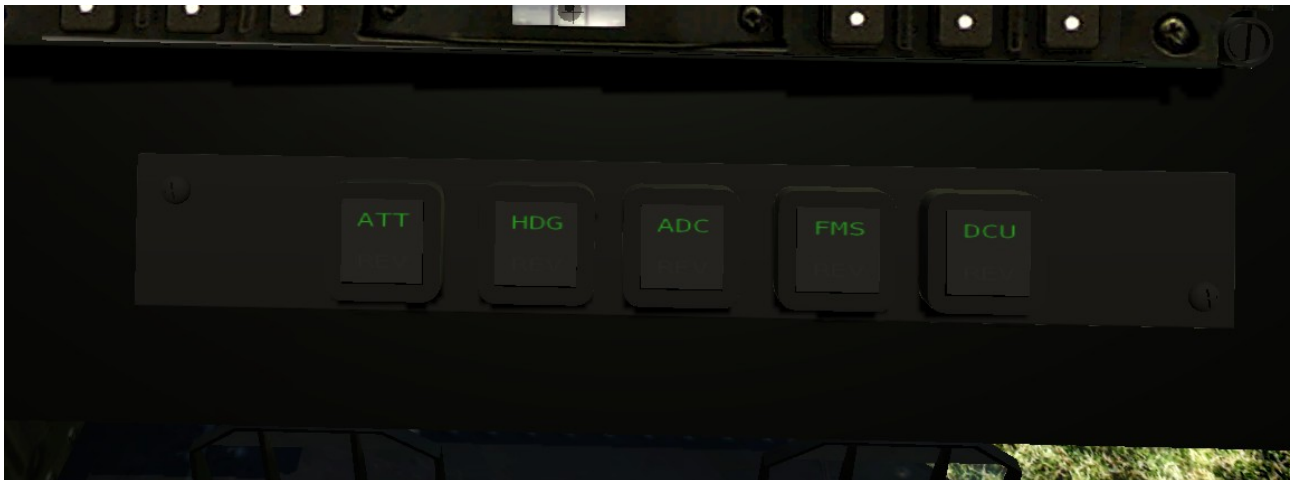




## Reversionary switch panel

The UH-60M has a number of redundant systems. The reversionary switch panel allows the pilot and co-pilot to switch between the primary and the secondary system. In the simulation these functions are limited by the available options provided by X-Plane.

At the moment it is possible to switch the ADC (air data) only. Depending of the switch setting, the MFDs show altitude (baro and radio), air speed, vertical speed and barometer setting from the “pilot” (green) or the “co-pilot” (amber) instruments.



# Weapons

## GUNS

Requirement: “GUNS” enabled in setup menu

Commands: uh60m/weapons/arm\_guns & uh60m/weapons/fire\_guns

Both “gunner doors” need to be open !

The guns are tied to the yaw axis. So they will follow the tail rotor pedals.

## CHAFF

Commands: uh60m/weapons/arm\_flares & uh60m/weapons/deploy\_chaff

## FLARE

Commands: uh60m/weapons/arm\_flares & uh60m/weapons/deploy\_flares

**Note:** weapons are disabled as long as we have WOW (weight on wheels)

See also: <https://www.youtube.com/watch?v=zKQfuxRcJ8I>



GUNS



CHAFF/FLARES



EOMS



## **Videos**

### **UH60M Videos playlist**

<https://www.youtube.com/watch?v=Muvl8dFNZUo&list=PLtspyUICsgSZDCGbs1DjPPdm96kQ2mA7F>

### **HowTo-clips playlist:**

<https://www.youtube.com/watch?v=AfriXZv96VE&list=PLtspyUICsgSaIVPmxsf2exlWePyU4dzOE>

**[Make sure you subscribe to this YouTube channel to not miss new videos](#)**

## **Missions**

see MISSIONS\_HOWTO.TXT in “missions” folder



## Datarefs and Commands

### Door commands

|                                |                          |
|--------------------------------|--------------------------|
| uh60m/doors/door_right_open    | Cockpit doors            |
| uh60m/doors/door_right_close   |                          |
| uh60m/doors/door_right_toggle  |                          |
| uh60m/doors/door_left_open     |                          |
| uh60m/doors/door_left_close    |                          |
| uh60m/doors/door_left_toggle   |                          |
| uh60m/doors/cargo_right_open   | Cargo doors              |
| uh60m/doors/cargo_right_close  |                          |
| uh60m/doors/cargo_right_toggle |                          |
| uh60m/doors/cargor_left_open   |                          |
| uh60m/doors/cargo_left_close   |                          |
| uh60m/doors/cargo_left_toggle  |                          |
| uh60m/doors/gun_right_open     | Gunner doors             |
| uh60m/doors/gun_right_close    |                          |
| uh60m/doors/gun_right_toggle   |                          |
| uh60m/doors/gun_left_open      |                          |
| uh60m/doors/gun_left_close     |                          |
| uh60m/doors/gun_left_toggle    |                          |
| uh60m/doors/avionics_open      | Avionics bay door        |
| uh60m/doors/avionics_close     |                          |
| uh60m/doors/avionics_toggle    |                          |
| uh60m/doors/hydraulics_open    | Hydraulics bay door      |
| uh60m/doors/hydraulics_close   |                          |
| uh60m/doors/hydraulics_toggle  |                          |
| uh60m/doors/engines_open       | Engine cowlings          |
| uh60m/doors/engines_close      |                          |
| uh60m/doors/engines_toggle     |                          |
| uh60m/doors/toggle_all         | Toggle all doors         |
| uh60m/doors/toggle_one         | Select door by dataref   |
| uh60m/doors/load_door_preset   | dref: uh60m/doors/preset |

### Light commands

|                            |                           |
|----------------------------|---------------------------|
| uh60m/lights/ldg_lt_on     | Landing light toggle      |
| uh60m/lights/srch_lt_on    | Search light toggle       |
| uh60m/lights/srch_lt_up    | Search light up           |
| uh60m/lights/srch_lt_dn    | Search light down         |
| uh60m/lights/srch_lt_lft   | Search light left         |
| uh60m/lights/srch_lt_rgt   | Search light right        |
| uh60m/lights/dome_blue     | Cargo dome light blue     |
| uh60m/lights/dome_white    | Cargo dome light white    |
| uh60m/lights/light_dn      | Dim cockpit lights        |
| uh60m/lights/light_up      | Brighten cockpit lights   |
| uh60m/lights/cockpit_flood | Bright flood light toggle |

### System and Engines commands

|                          |                               |
|--------------------------|-------------------------------|
| uh60m/engn/start_engine0 | Start engine 1                |
| uh60m/engn/start_engine1 | Start engine 2                |
| uh60m/engn/idle_stop0    | Release idle stop latch 1     |
| uh60m/engn/idle_stop1    | Release idle stop latch 2     |
| uh60m/sys/hover_hold     | en/disable hover hold         |
| uh60m/sys/level_hold     | en/disable level hold         |
| uh60m/sys/sas_boost      | en/disable SAS boost          |
| uh60m/sys/hstab_up       | Move stabilator manually up   |
| uh60m/sys/hstab_dn       | Move stabilator manually down |
| uh60m/sys/clock_sel      | Select clock/timer mode       |
| uh60m/sys/clock_ctrl     | start/stop/reset timer        |
| uh60m/sys/ftr            | Force Trim Release            |
| uh60m/caution_ack        | Acknowledge caution warnings  |
| uh60m/menu               | Open setup menu               |

### OPS commands

|                             |  |
|-----------------------------|--|
| uh60m/ops/winch_prep        |  |
| uh60m/ops/winch_ena         |  |
| uh60m/ops/winch_conn        |  |
| uh60m/ops/winch_up          |  |
| uh60m/ops/winch_down        |  |
| uh60m/ops/rescuer_hook_only |  |

|                             |                                   |
|-----------------------------|-----------------------------------|
| uh60m/ops/ipad_home         | Ipad home button                  |
| uh60m/ops/ipad_toggle       | Ipad show toggle                  |
| uh60m/ops/fuel_probe_out    | Extend fuel probe                 |
| uh60m/ops/fuel_probe_in     | Retract fuel probe                |
| uh60m/ops/fuel_probe_toggle | Toggle fuel probe                 |
| uh60m/ops/rope_out          | Show ropes                        |
| uh60m/ops/rope_in           | Hide ropes                        |
| uh60m/ops/drop_dude         | Drop seal                         |
| uh60m/ops/extract_seal      | Place Seal at OSM pos and set FMS |
| uh60m/ops/iff_emerg         | Sets transponder to 7700          |

#### Nav commands

|                      |                             |
|----------------------|-----------------------------|
| uh60m/nav/nav_src_up | Select next RNAV source     |
| uh60m/nav/nav_src_dn | Select previous RNAV source |

#### Weapon commands

|                             |                      |
|-----------------------------|----------------------|
| uh60m/weapons/arm_guns      | Arm GUNS             |
| uh60m/weapons/fire_guns     | Fire GUNS            |
| uh60m/weapons/arm_flares    | Arm CHAFF and FLARES |
| uh60m/weapons/deploy_flares | Deploy FLARES        |
| uh60m/weapons/deploy_chaff  | Deploy CHAFF         |

#### Camera commands

|                          |                             |
|--------------------------|-----------------------------|
| uh60m/cam/p1_select      | Select Pilot camera         |
| uh60m/cam/p2_select      | Select CoPilot camera       |
| uh60m/cam/c1_select      | Select CrewChief camera     |
| uh60m/cam/g1_select      | Select ground crew 1 camera |
| uh60m/cam/g2_select      | Select ground crew 2 camera |
| uh60m/cam/hoist_select   | Select hoist camera         |
| uh60m/cam/add            | Add position to path        |
| uh60m/cam/next           | Go to next path position    |
| uh60m/cam/reset          | Go to first path position   |
| uh60m/cam/clear          | Clear path                  |
| uh60m/cam/save           | Save paths to new.cam       |
| uh60m/cam/update         | Update current position     |
| uh60m/cam/load           | Load paths from default.cam |
| uh60m/cam/load_preflight | Load preflight path         |

**MFD commands**

|                     |                                |
|---------------------|--------------------------------|
| uh60m/mfd/cam_power | FLIR Camera power              |
| uh60m/mfd/cam_lock  | FLIR Camera locked to target   |
| uh60m/mfd/mfd1_pwr  | Power button on MFD1           |
| uh60m/mfd/mfd1_t1   | MFD1 top row button 1          |
| uh60m/mfd/mfd1_t2   | MFD1 top row button 2 ... 6    |
| uh60m/mfd/mfd1_r1   | MFD1 right row button 1        |
| uh60m/mfd/mfd1_r2   | MFD1 right row button 2 .... 6 |
| uh60m/mfd/mfd1_b1   | MFD1 bottom row button 1       |
| uh60m/mfd/mfd1_b2   | MFD1 bottom row button 2 ... 6 |
| uh60m/mfd/mfd1_l1   | MFD1 left row button 1         |
| uh60m/mfd/mfd1_l2   | MFD1 left row button 2 ... 6   |
| uh60m/mfd/q1up      | Queue 1 (yellow) msg box up    |
| uh60m/mfd/q1dn      | Queue 1 (yellow) msg box down  |
| uh60m/mfd/q2up      | Queue 2 (white) msg box up     |
| uh60m/mfd/q2dn      | Queue 2 (white) msg box down   |
| uh60m/mfd/mfd_inc   | Increment selected instrument  |
| uh60m/mfd/mfd_dec   | Decrement selected instrument  |
| uh60m/mfd/mfd_inc2  | Increment fast sel. Instr.     |
| uh60m/mfd/mfd_dec2  | Decrement fast sel. Instr.     |
| uh60m/mfd/mfd_push  | Push selected instrument       |
| uh60m/mfd/mfd1_s1   | MFD 1 select field 1           |
| uh60m/mfd/mfd1_s2   | MFD 1 select field 2           |
| uh60m/mfd/mfd1_s3   | MFD 1 select field 3           |
| uh60m/mfd/mfd1_s4   | MFD 1 select field 4           |
| uh60m/mfd/mfd1_s5   | MFD 1 select field 5           |
| uh60m/mfd/mfd1_s6   | MFD 1 select field 6           |

**FDCP commands**

|                    |                           |
|--------------------|---------------------------|
| uh60m/mfd/fdcp1d11 | FDCP 1 Display 1 button 1 |
| uh60m/mfd/fdcp1d12 | FDCP 1 Display 1 button 2 |
| uh60m/mfd/fdcp1d13 |                           |
| uh60m/mfd/fdcp1d21 |                           |
| uh60m/mfd/fdcp1d22 |                           |
| uh60m/mfd/fdcp1d23 |                           |
| uh60m/mfd/fdcp1d31 |                           |
| uh60m/mfd/fdcp1d32 |                           |



|                         |                           |
|-------------------------|---------------------------|
| uh60m/mfd/fdcp1d33      |                           |
| uh60m/mfd/fdcp1d41      |                           |
| uh60m/mfd/fdcp1d42      |                           |
| uh60m/mfd/fdcp1d51      |                           |
| uh60m/mfd/fdcp1d52      |                           |
| uh60m/mfd/fdcp2d33      |                           |
| uh60m/mfd/fdcp1_push1   | Push rotary 1             |
| uh60m/mfd/fdcp1_push2   | Push rotary 2             |
| uh60m/mfd/fdcp1_push3   | Push rotary 3             |
| uh60m/mfd/fdcp1_push4   | Push rotary 4             |
| uh60m/mfd/fdcp1_push5   | Push rotary 5             |
| uh60m/mfd/fdcp1_nxt_but | Select next button        |
| uh60m/mfd/fdcp1_prv_but | Select previous button    |
| uh60m/mfd/fdcp1_sel_but | Push selected button      |
| uh60m/mfd/fdcp1_nxt_psh | Select next rotary        |
| uh60m/mfd/fdcp1_prv_psh | Select previous rotary    |
| uh60m/mfd/fdcp1_sel_psh | Push selected rotary      |
| uh60m/mfd/fdcp1_rot_inc | Increment selected rotary |
| uh60m/mfd/fdcp1_rot_dec | Decrement selected rotary |

#### CDU commands

|                      |                 |
|----------------------|-----------------|
| uh60m/cdu/CDU1-popup | Open CDU1 popup |
| uh60m/cdu/CDU2-popup | Open CDU2 popup |
| uh60m/cdu/CDU1_L1    |                 |
| uh60m/cdu/CDU1_L2    |                 |
| uh60m/cdu/CDU1_L3    |                 |
| uh60m/cdu/CDU1_L4    |                 |
| uh60m/cdu/CDU1_L5    |                 |
| uh60m/cdu/CDU1_R1    |                 |
| uh60m/cdu/CDU1_R2    |                 |
| uh60m/cdu/CDU1_R3    |                 |
| uh60m/cdu/CDU1_R4    |                 |
| uh60m/cdu/CDU1_R5    |                 |
| uh60m/cdu/CDU1_CLR   |                 |
| uh60m/cdu/CDU1_COM   |                 |

|                     |              |
|---------------------|--------------|
| uh60m/cdu/CDU1_NAV  |              |
| uh60m/cdu/CDU1_FPN  |              |
| uh60m/cdu/CDU1_DAT  |              |
| uh60m/cdu/CDU1_INI  |              |
| uh60m/cdu/CDU1_XPDR |              |
| uh60m/cdu/CDU1_ZRO  |              |
| uh60m/cdu/CDU1_ENT  |              |
| uh60m/cdu/CDU1_NXT  |              |
| uh60m/cdu/CDU1_PRV  |              |
| uh60m/cdu/CDU1_DIM  |              |
| uh60m/cdu/CDU1_BRT  |              |
| uh60m/cdu/CDU1_N0   |              |
| uh60m/cdu/CDU1_N1   |              |
| uh60m/cdu/CDU1_N2   |              |
| uh60m/cdu/CDU1_N3   |              |
| uh60m/cdu/CDU1_N4   |              |
| uh60m/cdu/CDU1_N5   |              |
| uh60m/cdu/CDU1_N6   |              |
| uh60m/cdu/CDU1_N7   |              |
| uh60m/cdu/CDU1_N8   |              |
| uh60m/cdu/CDU1_N9   |              |
| uh60m/cdu/CDU1_NMIN |              |
| uh60m/cdu/CDU1_NDOT |              |
| uh60m/cdu/CDU1_LFT  | Cursor left  |
| uh60m/cdu/CDU1_RGT  | Cursor right |
| uh60m/cdu/CDU1_UP   | Cursor up    |
| uh60m/cdu/CDU1_DN   | Cursor down  |

#### **Mission commands**

|                        |                              |
|------------------------|------------------------------|
| uh60m/mission/start    | Start mission                |
| uh60m/mission/stop     | Stop/Hold mission            |
| uh60m/mission/load     | Load mission from file       |
| uh60m/mission/download | Download mission from server |
| uh60m/mission/save     |                              |
| uh60m/mission/reset    |                              |
| uh60m/mission/clear    |                              |

|                       |                                  |
|-----------------------|----------------------------------|
| uh60m/mission/add_wpt | Add current position to WPT list |
| uh60m/mission/skip    | Skip current wpt or task         |
| uh60m/mission/delete  |                                  |
| uh60m/mission/insert  |                                  |
| uh60m/mission/add     |                                  |

#### Airfuel commands

|                     |                              |
|---------------------|------------------------------|
| uh60m/ops/c130_show | Toggle aerial refueling C130 |
|---------------------|------------------------------|

## Datarefs

|                        |                                    |
|------------------------|------------------------------------|
| uh60m/cdu/mission_name | Name of selected mission file name |
| uh60m/door/preset      | Selected door preset               |
| uh60m/conf/probe       | Refuel probe installed             |
| uh60m/conf/probe_ext   | Refuel probe extended              |
| uh60m/conf/hoist       | Hoist winch installed              |
| uh60m/conf/hrst        | Rope mount installed               |
| uh60m/conf/doors       | Cockpit doors installed            |
| uh60m/conf/seats       | Cargo seats installed              |
| uh60m/conf/cargo       |                                    |
| uh60m/conf/guns        | Guns installed                     |
| uh60m/conf/pilots      | Pilots installed                   |
| uh60m/conf/flir        | FLIR cam installed                 |
| uh60m/conf/earplugs    | Earplug / Headset                  |
| uh60m/sys/sw/hstab     | HSTAB switch                       |
| uh60m/sys/sw/sas1      | SAS1 switch                        |
| uh60m/sys/sw/sas2      | SAS2 switch                        |
| uh60m/sys/sw/sas_boost | SAS boost switch                   |
| uh60m/sys/sw/tailwheel | Tailwheel lock switch              |
| uh60m/sys/sw/egi1      | EGI 1 switch                       |
| uh60m/sys/sw/egi2      | EGI 2 switch                       |
| uh60m/sys/sw/ralt_test | Radio altimeter test               |
| uh60m/sys/sw/hydr_bkup | Hydraulic backup switch            |
| uh60m/sys/sw/air_src   | Air source for starter             |

|                            |                              |
|----------------------------|------------------------------|
| uh60m/sys/sw/stby_inst     | Backup / Standby instrument  |
| uh60m/sys/sw/lamp_test     | Lamp test                    |
| uh60m/sys/sw/fire_test     | Fire test                    |
| uh60m/sys/hover_mode       | HSI Hover mode               |
| uh60m/sys/hover_hold       | Hover hold state             |
| uh60m/sys/level_hold       | Level hold state             |
| uh60m/sys/fps_on           | FPS / Heading-hold           |
|                            |                              |
| uh60m/cam/selected         | Selected camera              |
|                            |                              |
| uh60m/ops/winch_ena        | Winch enabled                |
|                            |                              |
| uh60m/weapons/flares_armed | CHAFF/FLARE armed            |
|                            |                              |
| uh60m/mfd/on               | Array/4 – MFD on             |
| uh60m/mfd/page             | Array/4 – MFD page           |
| uh60m/mfd/tgt_mode         |                              |
| uh60m/mfd/hsi_mode         |                              |
| uh60m/mfd/show_efis        |                              |
| uh60m/mfd/show_wxr         | Show EFIS weather radar      |
| uh60m/sys/CB               | Array/100 – circuit breakers |

#### functional CBs

CB\_hstab = 8  
 CB\_sas = 9  
 CB\_vhf1 = 12 → com1 & com2 power  
 CB\_tail\_lock = 13  
 CB\_master\_wrn = 15  
 CB\_egi1 = 34 → gps power  
 CB\_egi2 = 16 → gps2 power  
 CB\_hydr\_pump = 24